Contribution Journal

Student: John'a Matthews

For this project, I contributed by working with the Wine dataset and developing and evaluating the machine learning model. My main responsibility was to understand the dataset, prepare the information for the model, select features, train the model, and compare the results. Working on these parts helped me understand how the different steps of a machine learning project work together.

One of the main things I worked on was examining the features in the Wine dataset. The dataset included measurements such as alcohol, malic acid, flavanoids, color intensity, and proline. I looked at different combinations of three features and tested how changing the features affected the model's performance. This helped me see that the features used in a machine learning model can have an effect on the accuracy of the predictions.

I also contributed to preparing the data for the machine learning model. I worked with the dataset so that it could be divided into training and testing data. I then used Logistic Regression as a classification model. The goal was for the model to learn from the labeled wine examples and predict the correct wine class for new data. This connected directly to what we learned about supervised learning because the dataset already contained known classes.

Through my part of the project, I became more comfortable working with machine learning data and using Python libraries such as pandas and scikit-learn. I practiced preparing data, selecting features, splitting data into training and testing sets, training a Logistic Regression model, and evaluating the results.

I also learned the importance of testing different approaches instead of assuming that one set of features will always give the best results. Working with the Wine dataset made the concepts of supervised learning easier for me to understand because I could see the model use data to make predictions.